

HSC Mathematics Advanced - Applications of Calculus Worksheet

Motion, Rates, Volume, Exponential Models and Trig Modelling

Name: _____ Class: _____ Date: _____

How to use this worksheet

1. Read the mini content box before each section.
2. Attempt the HSC-style questions in the space provided.
3. Check your working with the full worked answers at the end.
4. For distance questions, always find where $v(t) = 0$ before integrating.

Core formulas you need to memorise

Question type	What to do	Common HSC trap
Acceleration $\rightarrow$ velocity	Integrate acceleration. Use $v(0)$ from the initial direction.	Left means negative velocity. Right means positive velocity.
Velocity $\rightarrow$ displacement	Integrate velocity normally: displacement = integral of $v(t)$.	Displacement can be negative.
Distance travelled	Find rest times by solving $v(t)=0$. Split intervals and add positive distances.	Do not just integrate $v(t)$ if the particle changes direction.
Position function	Integrate velocity and use $x(0)$ to find C.	At the origin means $x(0)=0$.
Rates of change	If given dV/dt , integrate to find V. If given V, differentiate to find rate.	Use the initial amount to find C.
Exponential models	Substitute information to find k. Differentiate for rate. Use logs for time.	Rate means derivative, not just value.
Trig models	Use amplitude, midline, period and solve trig equations/inequalities.	Check the correct time interval.

Section A - Motion from acceleration

Mini content

Acceleration is the derivative of velocity: $a = dv/dt$.

Velocity is the derivative of displacement: $v = dx/dt$.

If the particle is at the origin initially, use $x(0) = 0$.

If it is moving left initially, the initial velocity is negative.

Question 1 (7 marks)

A particle moves along the x-axis with acceleration $x'' = 10 \sin(2t)$. Initially, the particle is at the origin and moving to the left at 3 m/s.

(a) Show that the velocity is $x' = 2 - 5 \cos(2t)$.

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(b) Find the time when the particle first comes to rest.

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(c) Find the displacement in the first $\pi/2$ seconds.

[illegible]

(d) Find the distance travelled in the first $\pi/2$ seconds.

[illegible]

Question 2 (6 marks)

A particle moves along the x-axis with acceleration $x'' = 6 \cos(3t)$. Initially, the particle is at $x = 4$ and moving to the right at 1 m/s.

(a) Find the velocity of the particle.

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(b) Find the position function $x(t)$.
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(c) Find the displacement after $\pi/3$ seconds.
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Question 3 (6 marks)

A particle moves along the x-axis with acceleration $x'' = 4 \sin(t)$. Initially, the particle is at the origin and moving to the left at 2 m/s.

(a) Show that $x' = 2 - 4 \cos(t)$.
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(b) Find the time when the particle first comes to rest.
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(c) Find the position of the particle at that time.
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(d) Find the distance travelled in the first π seconds.
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Section B - Velocity, displacement and distance

Mini content

Displacement = integral of velocity over the interval.

Distance = total path travelled. Use absolute values or split at $v(t) = 0$.

To test direction, substitute values into velocity, not displacement.

Question 4 (6 marks)

The velocity of a particle is $v(t) = 3 - 4 \cos(t)$, for $0 \leq t \leq \pi$. Initially, the particle is at $x = 2$.

(a) Find the time when the particle first comes to rest.

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(b) Find the displacement from $t = 0$ to $t = \pi$.

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(c) Find the final position of the particle at $t = \pi$.

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(d) Find the total distance travelled from $t = 0$ to $t = \pi$.

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Question 5 (6 marks)

The velocity of a particle moving in a straight line is $v(t) = 3t^2 - 12t + 9$, for $0 \leq t \leq 5$.

(a) Find the times when the particle is at rest.

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(b) Find the displacement in the first 5 seconds.

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(c) Find the total distance travelled in the first 5 seconds.

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Question 6 (7 marks)

A particle moves along the x-axis with velocity $v(t) = 1 - 2 \cos(2t)$, for $0 \leq t \leq \pi/2$. Initially, $x = 0$.

(a) Find when the particle first comes to rest.

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(b) Find the displacement in the first $\pi/2$ seconds.

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(c) Find the distance travelled in the first $\pi/2$ seconds.

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Question 7 (6 marks)

A particle moves along a line with velocity $v(t) = 2 \sin(2t) - 1$, for $0 \leq t \leq \pi$.

(a) Find all times when the particle is at rest.

(b) Find the displacement from $t = 0$ to $t = \pi$.

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(c) Find the total distance travelled from $t = 0$ to $t = \pi$.

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Section C - Rates of change and volume

Mini content

If the question gives a rate like dV/dt , integrate to find the amount V .
Use the initial amount to find the constant C .
If the question asks when the rate is zero, solve $dV/dt = 0$ first.

Question 8 (3 marks)

Initially there are 250 L of water in a tank. Water flows into the tank at a rate $dV/dt = 250 - 7.5t$, where t is in hours. Find the volume in the tank when $dV/dt = 0$, correct to the nearest litre.

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Question 9 (4 marks)

A tank initially contains 120 L of water. Water flows in at a rate $dV/dt = 80 - 4t$, where t is in minutes.

(a) Find when the rate of increase becomes zero.

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(b) Find the volume in the tank at that time.

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Question 10 (5 marks)

Oil leaks from a container at a rate $dV/dt = 12e^{(-0.2t)}$, where V is the volume of oil lost in litres and t is in minutes.

(a) Find the amount of oil lost in the first 10 minutes.

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(b) Find the total amount of oil that will eventually leak out.

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Question 11 (4 marks)

The radius r cm of a circular oil spill is increasing at a rate $dr/dt = 0.4$ cm/s. Find the rate at which the area of the oil spill is increasing when $r = 12$ cm. Use $A = \pi r^2$.

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Section D - Exponential growth and decay

Mini content

Exponential models often look like $N = A(b)^{(kt)}$ or $N = Ae^{(kt)}$.

To find k , substitute a known value and use logarithms.

If the question asks for the rate after a time, differentiate the model first.

Question 12 (5 marks)

The number of concert tickets sold is modelled by $N = 1000(1.6)^{(kt)}$, where t is the number of weeks after tickets become available. After 2 weeks, the total number sold has doubled.

(a) Calculate k , leaving your answer in exact form.

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(b) At what rate was the number of tickets sold increasing after 4 weeks? Give your answer to the nearest ticket per week.

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Question 13 (5 marks)

A bacteria culture is modelled by $N = 500e^{(0.3t)}$, where t is in hours.

(a) Find the number of bacteria initially.

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(b) Find the rate at which the population is increasing after 4 hours.

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(c) Find the time when the population reaches 2000.

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Question 14 (6 marks)

Trout are released into a lake. The population is modelled by $P = A/(1 + 15e^{(-0.5t)})$, where t is in months. Initially, 250 trout were released.

(a) Show that $A = 4000$.

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(b) Find the range of P .

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(c) Find the rate at which the population is growing after 2 months, to the nearest trout per month.

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(d) Find how many whole months it takes for the population to exceed 3000.

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Section E - Trig modelling and area applications

Mini content

Trig models are used for periodic situations such as tides.

For $D = a \cos(bt) + c$: amplitude = $|a|$, midline = c , period = $2\pi/b$.

For area between curves, find intersections first, then top minus bottom.

Question 15 (6 marks)

The depth D metres of water over a sandbar is $D = 2 \cos((4\pi/25)t) + 3$, where t is hours after midnight. A boat can cross only when the depth is at least 4 metres.

(a) Find the maximum and minimum depth.

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The shaded logo is bounded by $y = \sin(x)$ and $y = x \sin(x)$ for $0 \leq x \leq \pi$.

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Question 17 (5 marks)

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Question 18 (5 marks)

The velocity of a particle is $v(t) = 4 \sin(2t) - 2$, for $0 \leq t \leq \pi$.

(a) Find all times when the particle is at rest.

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(b) Find the displacement in the first π seconds.

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(c) Find the total distance travelled in the first π seconds.

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Worked answers

Marking reminder

For full marks, show substitution of initial conditions, solve rest times exactly where possible, and clearly state units.
For distance questions, write either integral of $|v|$ or show position differences with absolute values.

Question 1

- (a) $v = \int 10 \sin(2t) dt = -5 \cos(2t) + C$. Since $v(0) = -3$, $-3 = -5 + C$, so $C = 2$. Therefore $v = 2 - 5 \cos(2t)$.
(b) $2 - 5 \cos(2t) = 0$, so $\cos(2t) = 2/5$. First time: $t = \frac{1}{2} \cos^{-1}(2/5)$.
(c) Displacement = $\int_0^{\pi/2} (2 - 5 \cos(2t)) dt = [2t - (5/2) \sin(2t)]_0^{\pi/2} = \pi$ m.
(d) Let $a = \frac{1}{2} \cos^{-1}(2/5)$. $x(t) = 2t - (5/2) \sin(2t)$. $x(a) = \cos^{-1}(2/5) - \sqrt{21}/2$. Distance = $-x(a) + (\pi - x(a)) = \pi - 2\cos^{-1}(2/5) + \sqrt{21}$ m.

Question 2

- (a) $v = \int 6 \cos(3t) dt = 2 \sin(3t) + C$. $v(0)=1$ gives $C=1$. $v = 1 + 2 \sin(3t)$.
(b) $x = \int (1 + 2 \sin(3t)) dt = t - (2/3)\cos(3t) + C$. $x(0)=4$ gives $C = 14/3$. $x(t) = t - (2/3)\cos(3t) + 14/3$.
(c) Displacement = $\int_0^{\pi/3} v dt = x(\pi/3) - x(0) = \pi/3 + 4/3$ m.

Question 3

- (a) $v = \int 4 \sin(t) dt = -4 \cos(t) + C$. $v(0)=-2$ gives $C=2$. $v=2-4\cos(t)$.
(b) $2 - 4\cos(t)=0$, $\cos(t)=1/2$, so $t=\pi/3$.
(c) $x = 2t - 4\sin(t)$. $x(\pi/3) = 2\pi/3 - 2\sqrt{3}$ m.
(d) Starts negative then positive after $\pi/3$. Distance = $|x(\pi/3) - x(0)| + |x(\pi) - x(\pi/3)| = (2\sqrt{3} - 2\pi/3) + (4\pi/3 + 2\sqrt{3}) = 4\sqrt{3} + 2\pi/3$ m.

Question 4

- (a) $3 - 4\cos(t)=0$, $\cos(t)=3/4$, so $t=\cos^{-1}(3/4)$.
(b) Displacement = $\int_0^{\pi} (3 - 4\cos t) dt = [3t - 4\sin t]_0^{\pi} = 3\pi$ m.
(c) Final position = initial position + displacement = $2 + 3\pi$.
(d) Let $a=\cos^{-1}(3/4)$. x change function from 0 is $s(t)=3t-4\sin t$. Since $v(0)<0$ then $v>0$ after a , distance = $-s(a) + (s(\pi)-s(a)) = 3\pi - 2(3a - 4\sin a)$. Since $\sin a = \sqrt{7}/4$, distance = $3\pi - 6\cos^{-1}(3/4) + 2\sqrt{7}$.

Question 5

- (a) $v=3(t-1)(t-3)$, so $t=1,3$.
(b) Displacement = $\int_0^5 (3t^2-12t+9) dt = [t^3 - 6t^2 + 9t]_0^5 = 20$ m.
(c) Position change function $s(t)=t^3-6t^2+9t$. $s(0)=0$, $s(1)=4$, $s(3)=0$, $s(5)=20$. Distance = $|4-0|+|0-4|+|20-0|=28$ m.

Question 6

- (a) $1 - 2\cos(2t)=0$, $\cos(2t)=1/2$. For $0 \leq t \leq \pi/2$, $t=\pi/6$.
(b) Displacement = $\int_0^{\pi/2} (t - \sin(2t)) dt = \pi/2$ m.
(c) Starts negative then positive after $\pi/6$. Distance = $\sqrt{3} + \pi/6$ m.

Question 7

- (a) $2\sin(2t)-1=0$, $\sin(2t)=1/2$. Since $0 \leq t \leq \pi$, $0 \leq 2t \leq 2\pi$. Thus $2t=\pi/6$ or $5\pi/6$, so $t=\pi/12$ or $5\pi/12$.
(b) Displacement = $\int_0^{\pi} (2\sin(2t)-1) dt = [-\cos(2t)-t]_0^{\pi} = -\pi$ m.
(c) Distance: split at $t=\pi/12$ and $t=5\pi/12$ using $s(t)=-\cos(2t)-t$. Total distance = $2\sqrt{3} + \pi/3$ m.

Question 8

Set $dV/dt=0$: $250 - 7.5t=0$, so $t=100/3$. $V = \int (250-7.5t) dt = 250t - 3.75t^2 + C$. $V(0)=250$, so $C=250$. $V(100/3)=4416.67$, so 4417 L.

Question 9

(a) $80 - 4t = 0$, so $t = 20$ minutes.

(b) $V = 80t - 2t^2 + C$, $C = 120$. $V(20) = 1600 - 800 + 120 = 920$ L.

Question 10

(a) Oil lost = $\int_0^{10} 12e^{-(0.2t)} dt = [-60e^{-(0.2t)}]_0^{10} = 60(1 - e^{-2})$ L.

(b) Total eventually = $\int_0^{\infty} 12e^{-(0.2t)} dt = 60$ L.

Question 11

$A = \pi r^2$. $dA/dt = 2\pi r dr/dt = 2\pi(12)(0.4) = 9.6\pi$ cm²/s.

Question 12

(a) $2000 = 1000(1.6)^{(2k)}$, so $2 = (1.6)^{(2k)}$. $k = \ln 2 / (2 \ln 1.6)$.

(b) $dN/dt = 1000(1.6)^{(kt)} k \ln(1.6)$. After 4 weeks, $N = 4000$, so $dN/dt = 4000k \ln(1.6) = 2000 \ln 2 = 1386$ tickets/week.

Question 13

(a) $N(0) = 500$.

(b) $dN/dt = 150e^{(0.3t)}$. At $t = 4$: $150e^{1.2}$, about 498 bacteria/hour.

(c) $2000 = 500e^{(0.3t)}$, so $4 = e^{(0.3t)}$. $t = \ln 4 / 0.3$, about 4.62 hours.

Question 14

(a) At $t = 0$, $250 = A/(1 + 15) = A/16$, so $A = 4000$.

(b) Since $t \geq 0$, P starts at 250 and approaches 4000 but never reaches it. Range: $250 \leq P < 4000$.

(c) $dP/dt = 30000e^{-(0.5t)} / (1 + 15e^{-(0.5t)})^2$. At $t = 2$: about 284 trout/month.

(d) $4000 / (1 + 15e^{-(0.5t)}) > 3000$ gives $e^{-(0.5t)} < 1/45$, so $t > 2 \ln 45 = 7.61$. Whole months: 8 months.

Question 15

(a) Maximum = 5 m, minimum = 1 m.

(b) Period = $2\pi / (4\pi/25) = 25/2 = 12.5$ hours.

(c) Need $2\cos((4\pi/25)t) + 3 \geq 4$, so $\cos((4\pi/25)t) \geq 1/2$. In first 25 hours: $0 \leq t \leq 25$. Let $u = (4\pi/25)t$, so $0 \leq u \leq 4\pi$. Solution: u in $[0, \pi/3] \cup [5\pi/3, 7\pi/3] \cup [11\pi/3, 4\pi]$. Convert $t = 25u/(4\pi)$: 0 to $25/12$ hours, $125/12$ to $175/12$ hours, and $275/12$ to 25 hours.

Question 16

(a) $\sin x = x \sin x \rightarrow \sin x(x-1) = 0$, so $x = 0, 1, \pi$.

(b) Area = $\int_0^1 (\sin x - x \sin x) dx + \int_1^\pi (x \sin x - \sin x) dx = \pi - 2\sin(1)$.

Question 17

Area = $\int_0^{10} 12x(10-x) dx = \int_0^{10} (120x - 12x^2) dx = [60x^2 - 4x^3]_0^{10} = 2000$ square units.

Question 18

(a) $4\sin(2t) - 2 = 0$, $\sin(2t) = 1/2$. Since $0 \leq 2t \leq 2\pi$, $2t = \pi/6$ or $5\pi/6$, so $t = \pi/12$ or $5\pi/12$.

(b) Displacement = $\int_0^\pi (4\sin(2t) - 2) dt = [-2\cos(2t) - 2t]_0^\pi = -2\pi$ m.

(c) Split at $t = \pi/12$ and $t = 5\pi/12$ using $s(t) = -2\cos(2t) - 2t$. Total distance = $4\sqrt{3} + 2\pi/3$ m.